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MILITARY SPECIFICATION

COVERS, SELF-SEALING, FUEL LINE, AIRCRAFT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for aircraft self-sealing fuel line covers, including self-sealing hoses.

1.2 Classification. The covers shall be of the following general types and protection levels, as specified (see 6.2):

Type I - For use with aluminum fuel lines

Type II - For use with steel fuel lines

Type III - For use with rigid nonmetallic fuel lines

Type IV - For use with flexible constructions (stainless-steel braid)

Type V - Integral with hose innertube

Protection level A - Cover is self-sealing against .30 caliber threat.

Protection level B - Cover is self-sealing against .50 caliber threat.

Protection level C - Cover is self-sealing against .60 caliber threat.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-S-735 Standard Test Fluids; Hydrocarbon

FSC 4720

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Aeronautical Systems Division, ASD/ENESS, Wright-Paterson AFB, OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-G-5572	Gasoline, Aviation: Grades 80/87, 100/130, 115/145
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-H-8794	Hose, Rubber, Hydraulic, Fuel and Oil Resistant
MIL-F-17874	Fuel Systems: Aircraft, Installation and Test of
MIL-F-38363	Fuel System, Aircraft, General Specification for

STANDARDS

Federal

FED-STD-601	Rubber: Sampling and Testing

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, publications, and drawings required by contractors in connection with specific procurement funtions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification provides for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. The materials used in the manufacture of the fuel line covers shall be suitable for the purpose and shall conform to applicable Government specifications. The covers shall not be highly flammable; the use of magnesium is prohibited. Where applicable Government specifications do not exist, contractor's specifications may be used provided that the contractor assumes full responsibility for the use thereof.

* 3.4 Design. The fuel line covers shall be designed for use with fuel lines selected in accordance with MIL-F-38363 for Air Force use. The fuel line covers shall be designed for use with fuel lines selected in accordance with MIL-F-17874 for Navy use. The fuel line covers shall be suitable for use with fuels conforming to MIL-G-5572 and MIL-T-5624. The weight of the covers shall be kept to a minimum and dimensions and dimensional tolerances shall be developed by the contractor or Government procuring activity.

3.5 Performance

3.5.1 Fungus resistance. The covers shall not deteriorate when exposed to fungus conditions as encountered in tropical climates.

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3.5.2 Humidity. The covers shall experience no deterioration or corrosion when exposed to a relative humidity of 95 percent including conditions wherein condensation takes place in the form of water or frost.

3.5.3 Salt atmosphere. The covers shall experience no deterioration or corrosion when exposed to salt-sea atmosphere as encountered around large bodies of salt water.

3.5.4 Temperature. The covers shall be operative over an ambient air temperature range of -55° to 107°C (-67° to +225°F).

3.5.5 Fuel resistance. The exterior surface of the covers shall be non-activating when placed in contact with jet fuels and jet fuel vapors.

3.5.6 Altitude. The covers shall perform satisfactorily at pressure altitudes from sea level to 75,000 feet.

3.5.7 Vibration. The covers shall be capable of withstanding vibration incident to service use without deterioration.

3.5.8 Leakage. The total leakage of the covers for the applicable protection level shall not exceed the limits as specified in 4.5.11.10 after any gunfire test round performed during the gunfire tests.

3.5.9 Gunfire resistance. The covers shall provide protection against gunfire as encountered in combat operations.

3.5.10 Functional performance of type V covers. The type V covers shall meet the following requirements.

3.5.10.1 Proof pressure. Each length of hose shall withstand a proof pressure of 300 pounds per square inch gage (psig) without indication of imperfections or leakage.

3.5.10.2 Reduction in diameter. The inside diameter (ID) of the hose shall not decrease to less than 90 percent of the minimum ID specified by the contractor or procuring activity.

3.5.10.3 Bending and vacuum. Application of a 28-inch mercury vacuum shall not result in more than an additional 10 percent reduction in outside diameter (OD) at any section for all sizes up to and including -24 and an additional 25 percent reduction for size -32 and larger.

3.5.10.4 Burst pressure. The hose shall not burst and there shall be no external leakage from the hose or end fittings below a burst pressure of 600 psig.

3.5.10.5 Barrier. The barrier shall be sufficient to preclude undue activation of the sealant.

3.6 Identification of product. Identification of covers shall be in accordance with applicable Government standards and shall include month and year of manufacture, type, and protection level.

3.7 Workmanship. Workmanship shall be in accordance with manufacturing practices covering this type of equipment. The sealant shall be free of

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foreign material, and the thickness shall agree with the approved construction within the established manufacturing tolerances as described in the manufacturer's specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of fuel line covers shall be classified as follows:

a. Preproduction tests (see 4.3)

b. Quality conformance tests (see 4.4).

4.3 Preproduction tests. The preproduction tests shall consist of phase I and phase II tests as specified in table I and shall be conducted on test specimens in the order listed therein as applicable to the protection level required. For the purposes of this specification, phase I and phase II tests are defined as follows:

a. Phase I preproduction tests are those tests performed on samples of the materials and construction to be used in the manufacture of the covers.

b. Phase II preproduction tests are those tests performed on phase I approved covered fuel lines, as specified by the procuring activity, for a particular aircraft installation.

4.3.1 Reports of tests

4.3.1.1 Phase I preproduction approval. Phase I preproduction test reports shall be prepared in accordance with MIL-STD-831. The reports shall be signed and approved by a responsible representative of the manufacturer or the laboratory in which the tests are conducted.

4.3.1.2 Preemptive approval. See 6.4

4.3.1.3 Phase II preproduction approval. Phase II preproduction test reports shall be prepared, signed, and approved as specified in 4.3.1.1.

4.4 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

The contractor shall furnish test reports showing quantitative results.

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TABLE I. Phase I and Phase II preproduction tests

Test	Test Specimen	Test Paragraph
PHASE I		
Fungus resistance	1	4.5.4
Humidity	1	4.5.5
Salt fog	1	4.5.6
Temperature -55°C (-67°F)	2	4.5.7
Temperature 107°C (+225°F)	2	4.5.7
Fuel resistance	1	4.5.8
Pressure	3-9	4.5.9
Vibration	3-9	4.5.10
Gunfire	3-9	4.5.11
Functional tests (Type V)	10-14	4.5.15
PHASE II		
Individual tests	All specimens	4.4.1
Gunfire	15-16	4.5.12
Adhesion	15-16	4.5.13

4.4.1 Individual tests. Each cover shall be subjected to the following tests as described under 4.5:

a. Examination of product (see 4.5.1)

b. Dimensional check (see 4.5.2)

c. Weight (see 4.5.3)

d. Proof pressure - type V only (see 4.5.15.1).

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* 4.4.2 Sampling plans and tests. Unless MIL-STD-105 or other standard sampling plans are specified by the Government procuring activity, the following sampling plans and tests shall apply.

4.4.2.1 Sampling plan A. The contractor shall submit two covers from each 150 production units to a gunfire test in accordance with the phase II gunfire test specified in 4.5.12.

4.4.2.2 Sampling plan B. The contractor shall submit one cover from each 150 production units to an adhesive bond test in accordance with 4.5.14.

4.4.2.3 Sampling plan C. The contractor shall submit one sample from each 100 type V production units to the burst pressure test in accordance with 4.5.15.4.

4.4.2.4 Rejection and retest. When an item from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain fully to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, the sampling tests shall be repeated. Covers rejected after retest shall not be resubmitted to the procuring activity without approval.

4.4.2.5 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of a sampling test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.5 Test methods

4.5.1 Examination of product. Each cover shall be carefully examined to determine conformance to all the requirements of this specification for which no specific tests are described and to determine conformance to approved manufacturer's material, fabrication, and inspection specifications.

4.5.2 Dimensional check. Each cover shall be subjected to a dimensional check to determine that all dimensions are within the dimensional tolerances established by the contractor or procuring activity.

4.5.3 Weight. Each cover shall be subjected to a weight inspection to determine that the weight requirements established by the contractor or Government procuring activity are met within a tolerance of ±4 percent.

4.5.4 Fungus resistance. The covers shall be subjected to a fungus test in accordance with MIL-STD-810, method 508.1, procedure I. There shall be no evidence of deterioration or corrosion.

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4.5.5 Humidity. The covers shall be subjected to a humidity test in accordance with MIL-STD-810, method 507.1, procedure I. There shall be no evidence of deterioration or corrosion.

4.5.6 Salt fog. The covers shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509.1, procedure I. There shall be no evidence of functional deterioration or corrosion.

4.5.7 Temperature. The covers shall be subjected to sustained (240 hours) operating temperature limits of -55°C (-67°F) and 107°C (+225°F). There shall be no evidence of deterioration or corrosion which could result in a reduction in self-sealing capability.

4.5.8 Fuel resistance. The covers shall be immersed for 1 hour in type III fluid at 57°C (135°F) conforming to TT-S-735. Any swelling, cover deterioration, or cover activation that will impair performance shall be cause for rejection.

* 4.5.9 Pressure. The covers shall be subjected to a low-pressure test in accordance with MIL-STD-810, method 500.1, procedure I. There shall be no evidence of deterioration.

* 4.5.10 Vibration. Prior to gunfire testing, the covers shall be subjected to a vibration test in accordance with MIL-STD-810, method 514.1, category b.1, b.2, or c, as applicable. Any indication of deterioration or corrosion shall be cause for rejection.

4.5.10.1 Vibration test setup. The test assembly, consisting of a 3-inch OD tube and cover for types I, II, and III covers or a 1-inch OD line for types IV and V, shall be rigidly mounted to a vibration table. Test assembly clamps shall be designed for aircraft use. The distance between the test assembly table supports shall be 30 inches. The test assembly shall be centered and supported 20 inches from the vibration table. The cover shall be 22 inches in length, and the major axis of the test assembly shall be perpendicular to the plane of vibration. The test assembly may be filled with JP-4 fuel conforming to MIL-T-5624 at 0 psig for the duration of this test.

4.5.11 Gunfire (phase I). The covers shall be tested by gunfire to determine their self-sealing capability. This test shall involve the use of five test specimens, and the covers shall be evaluated for 20 rounds. After each round the wound shall be plugged or clamped in such a manner as to eliminate all further leakage (see 6.5). A 6061-T6 aluminum tube with a wall thickness of .049 inch shall be used for type I cover testing, and a stainless steel tube (alloy 321) with a wall thickness of .035 inch shall be used for type II cover testing. Types III, IV, and V test specimens shall be as specified by the procuring activity. Armor-piercing ammunition shall be used for all gunfire testing except for protection level C tests (see note 1 to tables II, III, and IV). Projectiles shall strike the covers within a 1-1/2-inch band centered on the tube centerline.

4.5.11.1 Firing schedule. A normal temperature, low temperature, and high temperature gunfire test shall be conducted as specified in table II, III, and IV.

4.5.11.2 Test fluid. Type I fluid conforming to TT-S-735 shall be used for all gunfire testing. This fluid may be circulated during the gunfire testing.

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TABLE II. Gunfire schedule - protection level A 1/.

Test Specimen	Round	
No. 3 (Ambient temp)	1	.30 cal nontumbled
	2	.30 cal nontumbled
	3	.30 cal nontumbled
	4	.30 cal nontumbled
No. 4 (Ambient temp)	5	.30 cal nontumbled - 40 psig
	6	.30 cal nontumbled - 40 psig
	7	.30 cal nontumbled - 40 psig
	8	.30 cal nontumbled - 40 psig
No. 5 (Ambient temp)	9	.30 cal nontumbled
	10	.30 cal nontumbled
	11	.30 cal nontumbled
	12	.30 cal 30° obliquity
No. 6 (High temp)	13	.30 cal nontumbled
	14	.30 cal nontumbled
	15	.30 cal nontumbled
	16	.30 cal 45° obliquity
No. 7 (Low temp)	17	.30 cal nontumbled
	18	.30 cal nontumbled
	19	.30 cal nontumbled
	20	.30 cal nontumbled
No. 8 (Contingency)	1	.30 cal nontumbled - 40 psig
	2	.30 cal 30° obliquity
	3	.30 cal 45° obliquity
No. 9 (Information) 2/	1	.50 cal nontumbled
	2	.50 cal nontumbled
	3	.50 cal 30° obliquity
	4	.50 cal 45° obliquity

1/ Armor piercing ammunition, standard Government issue APM2 shall be used for gunfire tests.

2/ Information shots for this protection level shall be accomplished on a 2-inch OD fuel line.

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TABLE III. Gunfire schedule - protection level B 1/.

Test Specimen	Round	2/
No. 3 (Ambient temp)	1	.50 cal nontumbled - 2,100 ft/sec
	2	.50 cal nontumbled - Std
	3	.50 cal nontumbled - 2,100 ft/sec
	4	.50 cal nontumbled - Std
No. 4 (Ambient temp)	5	.50 cal nontumbled - Std
	6	.50 cal nontumbled - Std
	7	.50 cal nontumbled - 2,100 ft/sec
	8	.50 cal 30° obliquity - 2,100 ft/sec
No. 5 (Ambient temp)	9	.50 cal nontumbled - Std - 40 psig
	10	.50 cal nontumbled - Std - 40 psig
	11	.50 cal nontumbled - 2,100 ft/sec - 40 psig
	12	.50 cal nontumbled - 2,100 ft/sec - 40 psig
No. 6 (High temp)	13	.50 cal nontumbled - 2,100 ft/sec
	14	.50 cal nontumbled - 2,100 ft/sec
	15	.50 cal nontumbled - Std
	16	.50 cal 45° obliquity - Std
No. 7 (Low temp)	17	.50 cal nontumbled - 2,100 ft/sec
	18	.50 cal nontumbled - Std
	19	.50 cal nontumbled - 2,100 ft/sec
	20	.50 cal nontumbled - Std
No. 8 (Contingency)	1	.50 cal nontumbled - Std - 40 psig
	2	.50 cal 30° obliquity - 2,100 ft/sec
	3	.50 cal 45° obliquity - Std
No. 9 (Information) 3/	1	.30 cal tumbled - Std
	2	.30 cal tumbled - Std

1/ Armor-piercing ammunition, standard Government issue APM2 shall be used for all gunfire testing.

2/ The velocity of 2,100 ft/sec corresponds to a loading of 200 grains of 5010 powder.

3/ Information shots for this protection level shall be accomplished on a 3-inch OD fuel line and shall include tumbled rounds.

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TABLE IV. Gunfire schedule - protection level C 1/.

Test Specimen	Round	
No. 3 (Ambient Temp)	1	.60 cal nontumbled
	2	.60 cal nontumbled
	3	.60 cal nontumbled
	4	.60 cal nontumbled
No. 4 (Ambient temp)	5	.60 cal nontumbled
	6	.60 cal nontumbled
	7	.60 cal nontumbled
	8	.60 cal 45° obliquity
No. 5 (Ambient temp)	9	.30 cal tumbled
	10	.30 cal tumbled
	11	.30 cal tumbled
	12	.30 cal tumbled
No. 6 (High temp)	13	.60 cal nontumbled
	14	.60 cal nontumbled
	15	.60 cal nontumbled
	16	.60 cal nontumbled
No. 7 (Low temp)	17	.60 cal nontumbled
	18	.60 cal nontumbled
	19	.60 cal nontumbled
	20	.60 cal nontumbled
No. 8 (Contingency)	1	.60 cal nontumbled
	2	.60 cal nontumbled
	3	.60 cal nontumbled
No Information Shots		

1/ Cartridge cal 60 ball T80E2 ammunition downloaded to 446 grains of IMR4996 powder shall be used for this series of tests.

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4.5.11.3 Test setup. Test specimens shall be mounted 50 feet from the gun for all protection levels. Types I, II, and III covered test specimens in accordance with 4.5.11 shall be supported at 34 inches; types IV and V covered specimens in accordance with 4.5.11 shall be tested with a distance of 27 inches between supports. The covered fuel line shall be pressurized in accordance with 4.5.11.4, and the test setup shall include equipment to accurately measure leakage in cubic centimeters. Government approval of this equipment and test setup shall be required.

4.5.11.4 Pressure. The initial line pressure for gunfire tests shall be 30 ±3 psig except as specified in tables II and III.

4.5.11.5 Low temperature gunfire. The covers shall be subjected to a low temperature gunfire test as specified in tables II, III, and IV. The test fluid shall be at a bulk fluid temperature of -32°C (-25°F) or lower immediately preceding each gunfire test round. The exterior surface of the cover shall be at a temperature of -18°C (0°F) or lower immediately preceding each gunfire test round.

4.5.11.6 High temperature gunfire. The covers shall be subjected to a high temperature gunfire test as specified in tables II, III, and IV. The test fluid shall be at a bulk fluid temperature of 71°C (160°F) or higher immediately preceding each gunfire test round.

4.5.11.7 Oblique rounds. The covers shall be subjected to shots which are made at both 30° and 45° to the major axis of the covered fuel line in accordance with tables II, III, and IV. These rounds shall be evaluated using the same criteria as shots made at 90° to the covered fuel line.

4.5.11.8 Information shots. A specimen shall be tested for information purposes in accordance with table II and III. The results of this test shall be reported as a part of the evaluation. All information tests shall be accomplished at 30 ±3 psig.

* 4.5.11.9 Contingency rounds. A contingency specimen will be permitted in the form of test specimen No. 8 (see tables II, III, and IV). The indicated contingency rounds may be substituted for any evaluation round at the discretion of the manufacturer. Contingency rounds shall be subject to the same temperature requirements and gunfire evaluations as the evaluation rounds for which they are substituted. Contingency rounds may be used for raising a rating, obtaining damp seals, or passing previously failed rounds; only those contingency rounds (maximum of three) may be used to pass previously failed rounds. Details of substitutions of contingency rounds for evaluation rounds must be reported in the test report.

4.5.11.10 Leakage evaluation. Total leakage in excess of the following limits within 4 minutes immediately following any gunfire test round shall be cause for rejection of the round: 2,000 cubic centimeters (cc) for protection level A; 3,000 cc for protection level B; and 7,500 cc for protection level C. The requirements of this paragraph shall be independent of the gunfire evaluation requirements of 4.5.11.11. A rejected round (maximum of three) may be retested by use of a contingency round or rounds (see 4.5.11.9).

4.5.11.11 Gunfire evaluation. In consideration of the appropriate firing schedule, the covers shall be rated as specified in table V. Leakage as specified in table V shall be the total leakage from the test specimen for a

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period of 4 minutes immediately following each gunfire test round. The following minimum rating shall be accumulated from the 20 evaluation rounds for the applicable protection level: 80 points for protection levels A and B and 65 points for protection level C. A contingency round or rounds (see 4.5.11.9) may be used for improving ratings.

4.5.11.12 Damp seal evaluation. In addition to the gunfire requirements of 4.5.11.10 and 4.5.11.11, the following minimum number of damp seals shall be required to be accumulated from the 20 evaluation rounds for the applicable protection level: 24 damp seals for protection levels A and B and 14 damp seals for protection level C. This requirement for damp seals is to be considered as single holes (entrance or exit) rather than rounds. Damp seals (see 6.3) shall be accomplished within 2 minutes immediately following gunfire for ambient and high temperature gunfire tests; for low temperature gunfire tests, damp seals shall be accomplished within 4 minutes immediately following gunfire. A contingency round or rounds (see 4.5.11.9) may be employed to meet damp seal evaluation minimums.

4.5.12 Gunfire (phase II). The specified covered lines shall be tested by gunfire (see 6.5). Test conditions shall be 30 ±5 psig or as specified by the procuring activity at ambient temperature with covered lines mounted as specified in 4.5.11.2. Two test specimens shall be mounted 50 feet from the gun and either type I fluid conforming to TT-S-735 or JP-4 fuel conforming to MIL-T-5624 shall be used. Tests for the applicable protection level shall be as follows:

a. For protection level A, the test shall consist of three rounds into each test specimen. A minimum of seven damp seals shall be accomplished. Caliber .30 armor-piercing ammunition, standard Government issue APM2 shall be used for these gunfire tests.

b. For protection level B, the test shall consist of three rounds into each test specimen. A minimum of six damp seals shall be accomplished. Caliber .50 armor-piercing ammunition, standard Government issue APM2 shall be used for these gunfire tests.

c. For protection level C, the test shall consist of three rounds into each test specimen. A minimum of five damp seals shall be accomplished. Cartridge caliber .60 ball T80E2 ammunition downloaded to 446 grains of IMR4996 powder shall be used for this series of tests.

This requirement for damp seals should be construed as single holes (entrance or exit) rather than rounds. Failure to meet the number of damp seals shall be cause for rejection.

4.5.13 Adhesion (preproduction). Covers tested in accordance with 4.5.12 shall be subjected to an adhesion test in accordance with FED-STD-601, method 8031. The results of this test shall be reportd as the adhesion test acceptance value and shall be used as a basis for the quality conformance adhesion tests in 4.5.14.

4.5.14 Adhesion (quality conformance). Production samples shall be subjected to an adhesion test in accordance with FED-STD-601, method 8031 (see 4.4.2.2). A deviation of more than ±5.0 percent from the acceptance value determined in 4.5.13 shall be cause for rejection.

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TABLE V. Gunfire evaluation

Points	Leakage Results (cc)
Protection Level A	
5	0-150
4	150-300
3	300-450
2	450-600
1	600-750
0	750-2,000
Failure	2,000
Protection Level B	
5	0-150
4	150-300
3	300-450
2	450-600
1	600-750
0	750-3,000
Failure	3,000
Protection Level C	
5	0-400
4	400-800
3	800-1,500
2	1,500-3,000
1	3,000-5,000
0	5,000-7,500
Failure	7,500

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4.5.15 Type V functional tests. Type V covers shall be subjected to the following proof pressure, reduction in diameter, bending and vacuum, burst pressure, and barrier tests.

4.5.15.1 Proof pressure. Each length of hose and hose assembly shall be subjected to a proof pressure of 300 psig. The test shall be made with water. Pressure shall be held not less than 30 seconds and not more than 5 minutes. All hose shall be thoroughly dried after testing. There shall be no external leakage.

4.5.15.2 Reduction in diameter. Two 3-inch lengths of hose shall be oil aged in accordance with MIL-H-8794 and measured at each end at least one-half inch inside the hose from the end. The ID of the hose shall not decrease to less than 90 percent of the minimum ID specified by the contractor or the procuring activity in accordance with 4.5.2.

4.5.15.3 Bending and vacuum. The hose shall be tested in accordance with the bending and vacuum test of MIL-H-8794. Application of the 28-inch mercury vacuum shall not result in more than an additional 10-percent reduction in OD at any section for all sizes up to and including -24 and an additional 25-percent reduction for size -32 and larger.

4.5.15.4 Burst pressure. Two unaged hose assemblies shall be tested in accordance with the burst pressure test of MIL-H-8794. The test fluid shall be water. The hose shall not burst nor shall there be any external leakage from the hose or end fittings below a burst pressure of 600 psig.

4.5.15.5 Barrier test. Two assemblies shall be filled with type III test fluid conforming to TT-S-735, and a pressure of 30 psi shall be applied to the assembly. The fluid shall be maintained at a temperature of 57°C 3°C (135°F ±5°F) for a period of 30 days. Upon completion of the 30-day period, the hoses shall be examined and there shall be no unreasonable swelling, sealant activation, or material degradation. One sample shall be disected and the inside dimension shall be checked while the other sample shall be subjected to a burst pressure test in accordance with 4.5.15.4.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2), in accordance with MIL-STD-794.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2), in accordance with MIL-STD-794.

5.3 Marking. In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The covers are intended for use with aircraft fuel lines to protect the lines from damage.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type and protection level required (see 1.2)

c. Levels of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 Damp seal. A damp seal is a seal with no seepage, running, or visible dripping.

* 6.4 Preemptive approval. Preemptive approval is herein defined as approval of phase I preproduction tests conducted without contractual coverage. Reports of tests should be prepared in accordance with MIL-STD-831 and furnished in duplicate to the address specified in 6.4.1. These reports should be signed and approved by a responsible representative of the manufacturer or the laboratory in which the tests are conducted. Preemptive approvals may be accepted at the option of the procuring activity (prime contractor, government agency, et cetera) and shall be subject to verification by the procuring government activity.

6.4.1 Requests for information pertaining to the preproduction tests should be addressed to the Aeronautical Systems Division (ENFE), Wright-Patterson Air Force Base, Ohio 45433.

6.5 Damaged tubes. Where tubes are damaged beyond repair, a new test specimen may be employed for the remainder of the tests specified for that tube. This substitution is not to be confused with the contingency specimen, and details of substitutions must be reported.

6.6 Marginal indicia. The outside margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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